Reproducing A-MEM: A Cost-Aware Study of Memory Linking and Evolution

FTEC5660 — Agentic AI for Business and FinTech | Individual Reproducibility Assignment

Paper: Xu et al. (2025), A-MEM: Agentic Memory for LLM Agents | Code and evidence: github.com/alistairkung/A-mem-reproducibility

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1. Project summary and research questions

How much of A-MEM's reported advantage can we reproduce, and what can we attribute that advantage to? Xu et al. compare three ways to store a conversation: Neither memory action, Linking Only and Full A-MEM with linking and neighbour evolution [1]. Full leads on both reported metrics in all five question categories. The comparison leaves two questions open. Does updating older notes help on its own? What changes in quality accompany the additional model charges?

Part A. Reproducing the advantage. The Full implementation reproduced the frozen author workflow exactly in replay. On three preselected LoCoMo conversations, Full repeated the published category win on both metrics in three of five categories. Only adversarial questions repeated the entire three-way ordering. In my supplementary analysis of 558 questions, the same saved answers put Full first under historical F1 (32.69%) and Neither first under updated author F1 (39.00%).

Part B. Attributing the advantage. I added Evolution Only to test whether neighbour updates help temporal reasoning independently. On 89 temporal questions, enabling updates changed F1 by -2.44 percentage points without linking and $+2.67$ with it. Neither still had the highest temporal F1. Across all questions, Evolution Only scored below Neither under both scoring conventions and cost slightly more.

1.1 Cost-aware scope

The released LoCoMo sample [2] contains ten conversations, 5,882 turns and 1,986 questions. I selected three conversations before inspecting performance, stratifying by length with a fixed seed (42). Crossing them with four configurations required twelve independent stochastic memory builds, down from forty for the full sample. Each configuration answered the same 558 questions. The design gives one build per conversation and configuration.

Conversation	Turns	Questions
conv-44	675	158
conv-49	509	196
conv-50	568	204
Total per condition	1,752	558

Table 1: Preselected LoCoMo evaluation population (558 questions per configuration).

2. A-MEM and experimental setup

2.1 Memory construction and retrieval

For each conversational turn, an LLM stores the original content with generated keywords, context and tags. MiniLM embeddings identify up to five nearby memories. A shared LLM decision proposes two actions. *Strengthen* adds outgoing links and may revise tags on the new note. *Update neighbour* can change contexts and tags on older notes while preserving their original content. After insertion, the note is available for answering questions. Answer retrieval starts with ten similar notes and can follow links to additional entries.

Neither still creates notes and metadata, retrieves memories and requests the shared decision whenever the frozen routine calls for it. The four configurations vary which proposed actions may change memory. All incur the cost of generating shared decisions.

2.2 Frozen runtime and reproducibility

The frozen author implementation [3] was fixed to commit f9a173d. The accepted GPT-4o-mini campaign used openai/gpt-4o-mini-2024-07-18 via OpenRouter, with OpenAI pinned as the upstream provider. Each cell used separate memory, retrieval-index and response-cache state. Prompts and native output contracts were fixed; temperature was 0.7 except for category-5 answers at 0.5. Construction neighbours were k=5; answer retrieval began at k=10. A 4,096-token limit applied to evolution decisions and 1,000 tokens to other stages. The recorded Python environment used Python 3.11.17 for publication reproducibility.

Deterministic replay used the archived DeepSeek responses from conv-50. The gated Full routine matched the frozen author routine at 568 memory insertion states, 1,544 requests and 204 answers. Separate tests verified that disabled actions left prohibited fields untouched and that per-condition memory stores stayed independent (Section 4). For the GPT-4o-mini comparison, each configuration built fresh memories [4, 5].

3. Reproduction target and evaluation

3.1 The published comparison

Table 3 of Xu et al. reports F1 and BLEU-1 for each question category [1]. Within every category both scores rise from Neither to Linking Only to Full. I reconstructed Neither and Linking Only alongside Full on our three selected conversations and compared category scores under historical evaluation (Section 5.1). I then assessed how the authors' updated reference changed our own question-weighted comparison (Section 5.2).

3.2 What the metrics mean

The released F1 scorer treats words as sets. Precision measures the share of distinct predicted words that appear in the reference, recall the share of reference words recovered. F1 is their harmonic mean. BLEU-1 measures unigram overlap and penalises short predictions. A valid paraphrase can score poorly on either metric. I score each question and take equal-weight means over the stated population.

3.3 What the five question categories test

Category	Questions	What the question requires
Multi-hop	99	Combine facts scattered across multiple conversation turns.
Temporal	89	Reason about dates, sequences and changes over time.
Open-domain	27	Apply general knowledge or contextual understanding to conversation evidence.
Single-hop	222	Retrieve a directly stated conversational fact.
Adversarial	121	Recognise when the conversation does not supply the requested information.

Table 2: The five LoCoMo question categories in the selected 558-question population.

3.4 Historical and updated author evaluation

The historical evaluator used the supplied distractor as the Category 5 reference. On 22 October 2026, before the study protocol was frozen, the authors changed the default to "Not mentioned in the conversation" and kept `--legacy_c5_reference` for historical scoring [6]. Our initial specification (D11) designated updated author evaluation as primary and historical evaluation as secondary for the paper comparison.

Our T4 implementation applies the same author F1 and BLEU-1 functions to each saved prediction twice. Only the reference for the 121 adversarial questions changes. A separate binary score records whether the model identified missing information. That count stays fixed when the overlap reference changes.

Saved response	Historical F1 / BLEU-1	Updated F1 / BLEU-1
Audrey (Full A-MEM), abstained	18.18 / 16.37	100 / 100
Dave (Evolution Only), chose distractor	100 / 100	0 / 0

Table 3: Two verified Category 5 predictions from the accepted GPT-4o-mini experiment. Each row scores one saved answer under both references (percent).

Audrey described a pet-bonding workshop. Asked why she had signed up for a car maintenance workshop, Full A-MEM answered "Not mentioned in the conversation". Only the word "the" overlaps the historical reference "Strengthen the bond with her pets". Dave described restoring old cars. Asked what he found satisfying about destroying them, Evolution Only gave the distractor verbatim: "Transforming something old and beat-up into something beautiful". The binary abstention scores are 1 for Audrey's question and 0 for Dave's under either view. The five-example audit gives the derivations for conv-44:qa:126 and conv-50:qa:183, respectively.

4. Modification, experimental wrapper and verification

4.1 Completing the factorial design

Condition	Strengthen new note	Update existing neighbours
Neither	OFF	OFF
Linking Only	ON	OFF
Evolution Only (added)	OFF	ON
Full A-MEM	ON	ON

Table 4: Two independently gated action bundles. The shared LLM evolution decision remains in all four conditions.

Evolution Only completes the 2×2 design. I expected that updating older notes might help with dates, sequences and changes over time. This motivated the temporal comparison, although I did not preregister a temporal endpoint. Linking might also help multi-hop questions that draw on facts from different turns.

4.2 Wrapper implementation and validation

The wrapper selects the condition, routes calls, isolates state and records evidence (Figure 1). In the frozen method's Python abstract syntax tree, I gated only its two existing actions (Listing 1). Note insertion, retrieval and consolidation retain their original behaviour.

```
gates = action_configuration(condition)
tree = ast.parse(verified_author_bytes(Path(root), 'memory_layer.py'))
cls = next(n for n in tree.body if isinstance(n, ast.ClassDef)
           and n.name == 'AgenticMemorySystem')
method = copy.deepcopy(next(n for n in cls.body
                    if isinstance(n, ast.FunctionDef) and n.name == '_evolve'))
for node in ast.walk(method):
    if isinstance(node, ast.If) and isinstance(node.test, ast.Compare):
        test = node.test
        if (isinstance(test.left, ast.Name) and test.left.id == 'action'
                and len(test.comparators) == 1
                and isinstance(test.comparators[0], ast.Constant)
                and test.comparators[0].value in gates):
            action = test.comparators[0].value
            node.test = ast.copy_location(
                ast.BoolOp(op=ast.And(), values=[test, ast.Constant(gates[action])]), test)
```

Listing 1: Abridged excerpt from amem_repro/actions.py, commit 8485673. The excerpt omits the equality-operator check, validation and compilation.

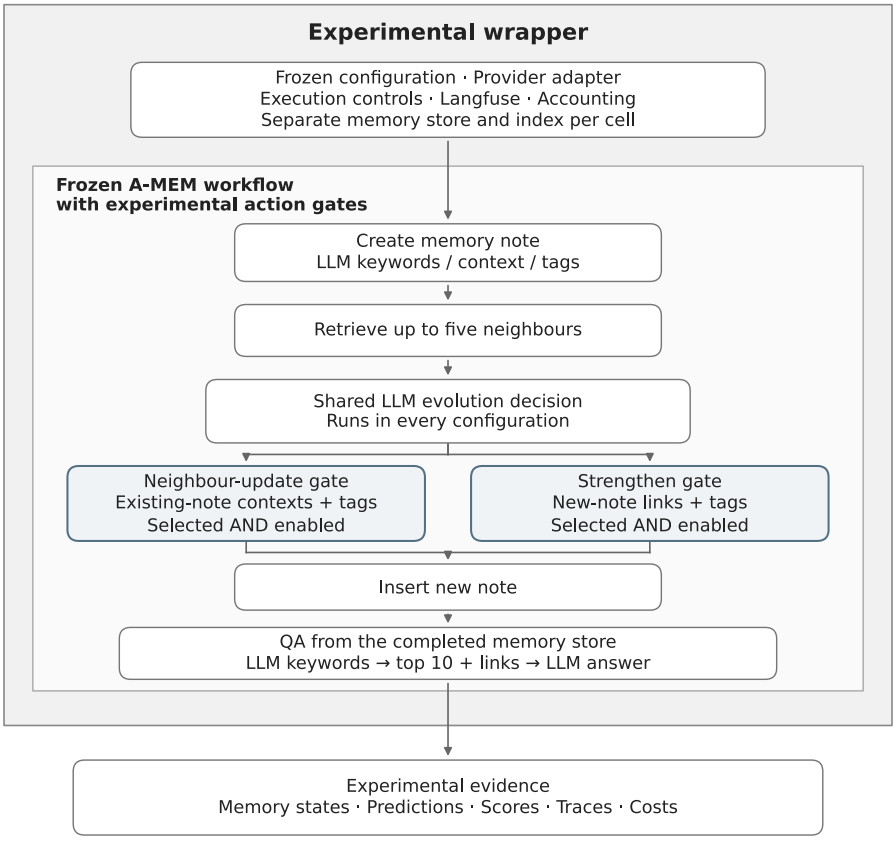


Figure 1: The experimental wrapper controls execution, tracing and accounting. The frozen A-MEM workflow supplies the shared decision and applies each selected action when its gate permits it.

4.3 Tracing and financial reconciliation

Langfuse traced generations, memory actions, retrieval and scores. Local journals and scientific records were authoritative. The controller reserved a maximum charge before transmission and settled using generation ID, route, native tokens and cost. Missing usage retained the full reservation.

5. Results and interpretation

5.1 Part A: Which published category results recur?

Configuration	Source	Multi-hop	Temporal	Open-domain	Single-hop	Adversarial
Neither	Paper	9.65/7.09	24.55/19.48	7.77/6.70	13.28/10.30	15.32/18.02
	Ours	16.86/11.94	**36.61/32.80**	14.37/14.15	34.82/28.72	30.66/29.90
Linking Only	Paper	21.35/15.13	31.24/27.31	10.13/10.85	39.17/34.70	44.16/45.33
	Ours	**20.25/15.00**	31.62/28.32	12.78/12.86	33.75/28.18	34.70/34.22
Full A-MEM	Paper	**27.02/20.09**	**45.85/36.67**	**12.14/12.00**	**44.65/37.06**	**50.03/49.47**
	Ours	17.54/11.29	34.29/30.13	**15.92/16.41**	**35.94/30.40**	**41.70/41.24**

Table 5: Table 3 of Xu et al. [1] and our historical reconstruction [5]. Each pair is F1/BLEU-1 (%). Bold marks each source's category winner on both metrics. Our sample sizes are 99, 89, 27, 222 and 121, respectively.

Full again won both metrics in open-domain, single-hop and adversarial. Only adversarial retained the complete ordering. On single-hop, Full kept first place and Neither overtook Linking Only for second. Linking Only won multi-hop and Neither won temporal.

5.2 How the reference changes our aggregate ranking

Configuration	Historical F1	Hist. BLEU-1	Updated F1	Updated BLEU-1	Adversarial correct
Neither	30.03	25.95	39.00	35.01	86/121
Linking Only	30.21	26.43	37.34	33.57	80/121
Evolution Only	29.29	25.69	35.86	32.29	79/121
Full A-MEM	32.69	28.64	36.91	32.81	71/121

Table 6: Question-weighted percentages on all 558 matched questions per configuration; two reference policies applied to the same saved predictions.

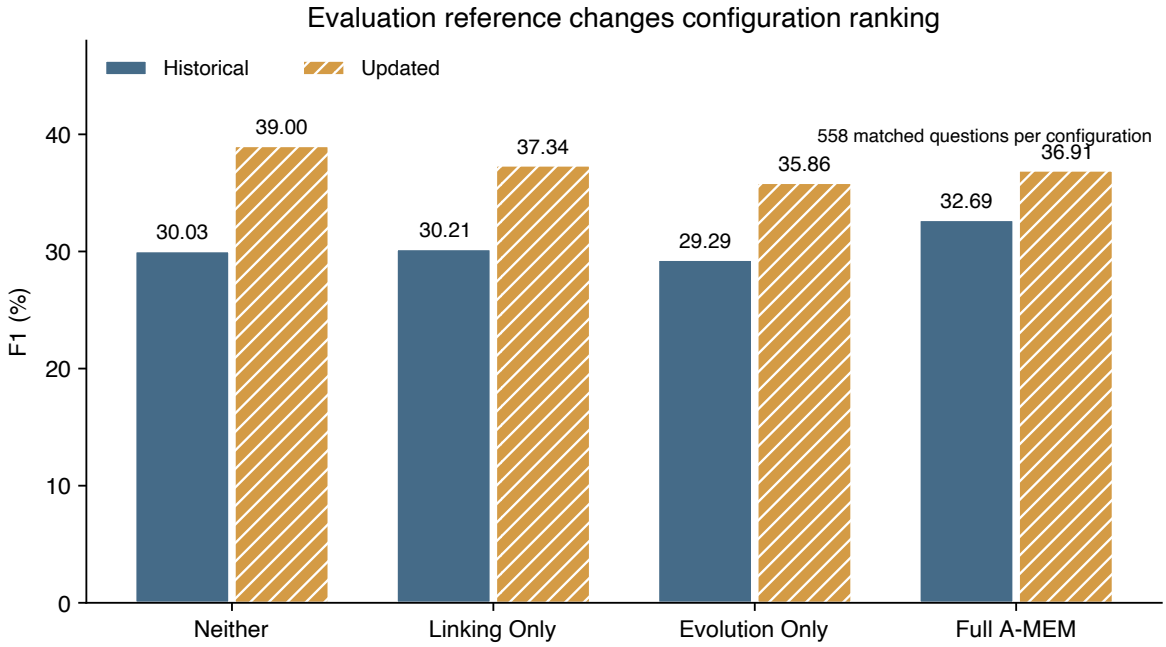


Figure 2: Across 558 matched questions per configuration, the historical distractor puts Full first (32.69% F1). Scoring the same answers against the no-information reference puts Neither first (39.00%). Only the 121 Category 5 references change.

The paper's Table 3 reports category scores only. Our supplementary aggregate weights each of the 558 questions equally. Under historical F1, Full led at 32.69%. Linking Only exceeded Neither by 0.18 points, Full exceeded Linking Only by 2.49, and Evolution Only fell 0.74 below Neither. Under updated author evaluation, Neither led on F1 (39.00%) and BLEU-1 (35.01%) and recognised missing information most frequently (86/121). Full abstained appropriately on 71/121.

For each configuration, all 437 questions in Categories 1–4 had identical scores under both policies. The other 121 questions had the same predictions and different references. This pairing attributes the change in the aggregate winner to rescoring those answers. To ask what the memory actions contributed, I next compare the independently built configurations under the planned primary metric.

5.3 Part B: Does neighbour evolution help temporal reasoning?

Evolution Only scored 34.17% temporal F1 against 36.61% for Neither (-2.44 points). With linking enabled, adding evolution raised temporal F1 from 31.62% to 34.29% ($+2.67$ points). The $+5.11$-point temporal interaction describes that change in effect. Neither still had the highest absolute temporal F1. Conv-49 differed from the pooled result for evolution alone. On its 33 temporal questions, Evolution Only exceeded Neither by 5.58 points. With one build per cell, conversation differences and stochastic variation cannot be separated.

Population	Evolution Only − Neither	Full − Linking Only	Interaction
Temporal (89)	-2.44 / -1.53	$+2.67$ / $+1.81$	$+5.11$ / $+3.33$
All questions (558)	-3.15 / -2.72	-0.43 / -0.76	$+2.72$ / $+1.96$

Table 7: Observed evolution contrasts under updated author evaluation. Each pair is the difference in F1/BLEU-1 (percentage points), calculated from unrounded question scores. Interaction subtracts the evolution contrast without linking from the contrast with linking.

Across all 558 questions, enabling evolution yielded lower updated F1 both without linking (-3.15 points) and with it (-0.43). The interaction was $+2.72$ points overall, positive in conv-44 and conv-50 and negative in conv-49. Evolution Only scored below Neither in all three conversations overall and in all five pooled categories. The overall changes from enabling linking were -1.66 F1 / -1.44 BLEU-1 points without evolution and $+1.06$ / $+0.52$ with evolution.

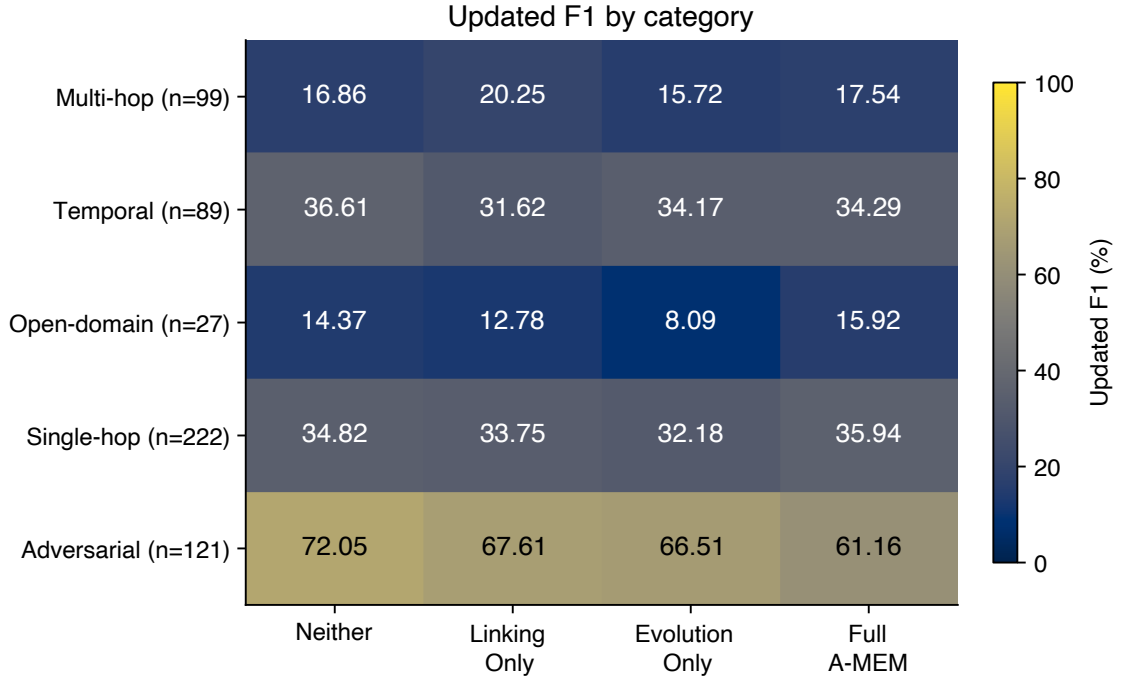


Figure 3: Updated F1 (%) on the five categories. The 89 temporal questions address the initial evolution hypothesis. Only the adversarial row changes when the historical reference is used.

Linking Only had the highest multi-hop F1, exceeding Neither by 3.40 points (20.25% versus 16.86%, rounded). It also retrieved more entries and used more answer input tokens (Table 8). These simultaneous differences leave the explanation for the score change open. Full led on open-domain and single-hop F1. The open-domain interaction was large ($+9.41$ points) but rested on just 27 questions. Neither led on updated adversarial F1 (72.05%), while the historical distractor gave Full the highest adversarial F1 (41.70%).

5.4 Quality and model charges across the four configurations

Following Professor Nguyen's Pareto approach, I compare each configuration's aggregate F1 with its identified model charges. A configuration is dominated if another costs no more and scores at least as well, with a strict advantage on at least one measure.

Configuration	Retrieved / QA	Answer input tokens	Model USD	Updated F1
Neither	10.00	747,640	$0.95045	39.00
Linking Only	20.15	1,511,478	$1.07939	37.34
Evolution Only	10.00	749,026	$0.95333	35.86
Full A-MEM	19.99	1,503,791	$1.08084	36.91

Table 8: Matched 558-question scientific inference costs from accepted builds only; excludes failed attempts, held reservations and Langfuse fees.

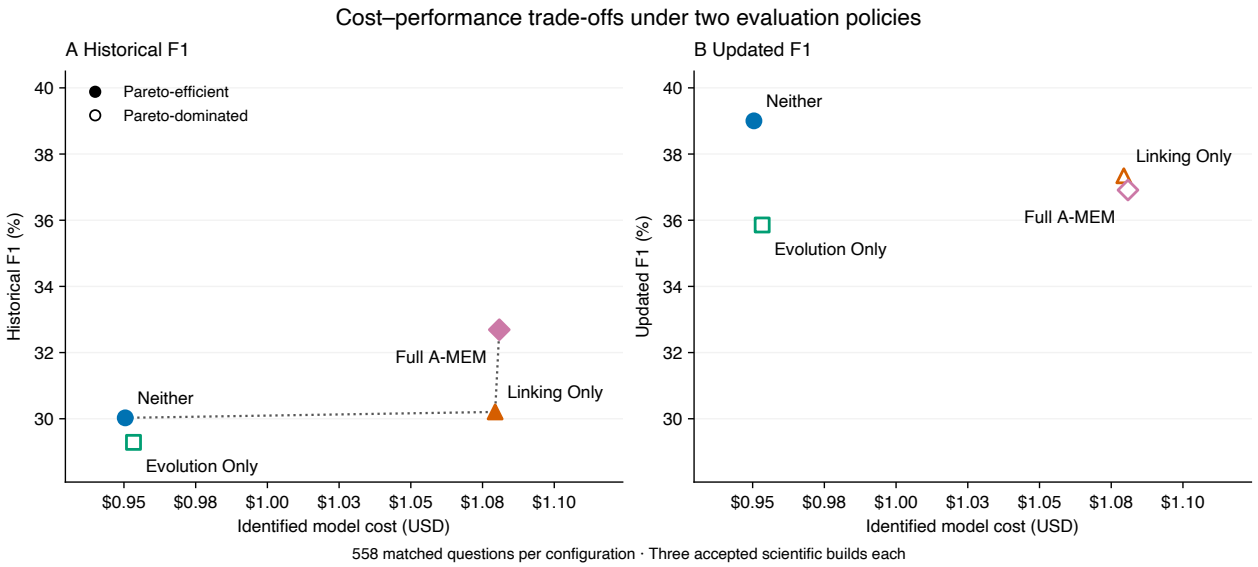


Figure 4: Cost–performance frontiers under historical and updated F1 (558 matched questions). Historical scores place Neither, Linking Only and Full A-MEM on the observed frontier; updated scores place only Neither on it. Costs include identified model charges from accepted scientific builds, excluding earlier attempts and Langfuse fees.

Across the three accepted builds per configuration, Linking Only increased mean retrieval from 10.00 to 20.15 entries per question and native answer input from 747,640 to 1,511,478 tokens. Its identified model charge rose from $0.95045 to $1.07939. Evolution Only incurred just $0.00288 more than Neither ($0.95333 versus $0.95045), with lower aggregate F1 under both policies. Neither therefore dominated Evolution Only on both frontiers. All configurations paid for the shared decision. These realised condition totals include the shared overhead and variation in generated tokens. They do not isolate the price of an individual mutation.

I reconciled identified charges against OpenRouter generation records, checking identities, upstream route, tokens and cost independently against the local ledger. Summed charges also matched wallet and API-key movements. All 1,666 generations in the final clean cell reconciled. The frontier uses only the twelve accepted builds. Failed attempts, held reservations and Langfuse fees appear in the campaign accounting in Section 6.

On the historical frontier, Neither, Linking Only and Full had successively higher charges and observed F1. Under updated F1, Neither alone was Pareto-efficient: it had both the highest aggregate F1 and the lowest identified model cost. The frontier applies to this model, these three conversations and the observed prices.

6. Debug diary and methodological reflection

6.1 The unplanned DeepSeek detour

The frontier counts charges from the accepted GPT-4o-mini builds. It leaves out the work that preceded them. I first chose DeepSeek because I already had credit and expected an OpenAI-compatible API to be straightforward to substitute. I encountered type mismatches in structured outputs, malformed JSON, output-limit failures and transport interruptions. My instinct was to solve each obstacle through more engineering: strict contract checks, checkpoint recovery, financial reservations and Langfuse instrumentation. These controls proved useful, but they also consumed time that should have been directed at the A-MEM mechanisms.

Three DeepSeek Flash Full A-MEM lineages completed in Degraded status. Table 9 pools their saved scores by question. On the same 558 questions, DeepSeek had higher observed F1 and BLEU-1 than GPT-4o-mini Full A-MEM under both policies. This comparison was exploratory. Each model contributed one build per conversation, and provider routes, generation caps, compatibility handling and recovery histories differed. I had spent more time porting the model than testing the memory mechanisms. I returned to GPT-4o-mini to follow the paper's model choice more closely in the primary factorial study.

Conversation	QA	Updated F1	Updated BLEU-1	Hist. F1	Hist. BLEU-1
conv-44	158	54.06	48.37	42.05	36.24
conv-49	196	49.63	46.31	43.22	39.95
conv-50	204	48.22	43.53	41.74	37.19
Pooled (derived)	558	50.37	45.88	42.35	37.89

Table 9: Exploratory DeepSeek Flash Full A-MEM results (scores in %). Three completed Degraded lineages; the pooled row is a derived 558-question aggregate, not part of the primary GPT-4o-mini experiment.

6.2 Provider failures and the final clean run

The final Linking Only / conv-44 cell exposed a second problem: execution infrastructure could affect the author's fallback behaviour. The first run disconnected during evolution; the special financial stop escaped the author's ordinary exception handler, preventing an unevolved note from being inserted. A reviewed recovery completed all 675 memories, but a second disconnect stopped answer generation after 87 of 158 questions. A fresh replacement then stopped on an HTTP 200 response whose generation was marked erroneous. In each case we preserved the failed artifacts rather than silently altering outputs or combining stochastic memory states.

I ultimately chose a final fresh run with broader, predeclared and financially bounded recovery rules while retaining unchanged scientific settings. The successful run completed 675 memories and 158 questions through 1,666 first-transmission model calls, with zero retries. Independent checks reproduced embeddings, retrieval rankings, scores, Langfuse experiment associations and provider charges. The new Linking Only cell was accepted as Clean, and all earlier failures remained available for audit.

Conservative accounting and provenance were necessary, but overly narrow stop rules and repeated full-history reconciliation made the experiment fragile and slow. A better design would combine immutable scientific state with bounded, failure-triggered retries and incremental financial reconciliation. Those choices would have preserved more of the project time for the mechanism comparison.

There was also an important difference between scientific inference cost and the cost of conducting research. The twelve accepted GPT-4o-mini builds incurred approximately $4.06 in identified model charges. Campaign-wide identified model charges, including earlier failed attempts and compatibility checks, totalled approximately $4.70. Broader conservative campaign exposure, including held reservations and a conservative upper bound for Langfuse service, reached $10.46. Only the accepted-build model charges belong in Section 5's like-for-like Pareto comparison.

The final Clean cell yielded a complete comparison without reconstructed memories, answer imputation or cross-run mixing. In a future study I would budget time for observability and accounting from the outset, alongside the time and credit for independent memory builds.

7. Limitations, conclusion and reproducibility evidence

7.1 Part A: How much of the published advantage recurs?

Deterministic Full replay matched the frozen author workflow under the archived responses. The stochastic experiment addressed a different question. Full again won both historical metrics in three of the five categories, while only adversarial questions retained the paper's entire ordering. On the 558-question supplementary comparison, historical scoring put Neither (30.03%), Linking Only (30.21%) and Full (32.69%) in ascending F1 order. Updated author scoring put Neither first (39.00%) after changing the reference for 121 questions. The answers were identical across the two evaluations.

7.2 Part B: What can we attribute to neighbour evolution?

Neighbour updates alone gave lower temporal F1 than Neither. Adding the same kind of updates to Linking Only gave higher temporal F1, a difference of $+5.11$ points between the two contrasts. Across all questions both contrasts were negative, with a $+2.72$-point interaction. Neither dominated Evolution Only on aggregate cost and F1 under both policies. Linking Only's multi-hop gain accompanied a larger retrieval set and more answer tokens.

7.3 Limits of inference

The paper's exact questions and retrieval settings for each Table 3 entry are unverified. Here the population was three of the ten released conversations and initial retrieval depth was ten. Each conversation and configuration had one stochastic build. We cannot estimate within-cell variance, separate conversation differences from run variation, or infer statistical significance. Strengthening changes links and tags together, and evolution changes older contexts and tags together. Repeating builds, adding conversations and independently varying retrieval depth and these individual operations would test how far the observed patterns extend.

7.4 Sources and development evidence

Xu et al. (2025), A-MEM: Agentic Memory for LLM Agents, arXiv:2502.12110 (v11); original implementation WujiangXu/A-mem (frozen f9a173d); LoCoMo benchmark data distributed with the release. Source, results, commit history and execution instructions are available at github.com/alistairkung/A-mem-reproducibility.

Accepted twelve-cell scientific evidence: 8485673183d7b8018c841082d075b21158002700. The repository archives source and dataset fingerprints, original question IDs, model responses, memory checkpoints and question-level scores. The publication generator pins its scientific inputs to this revision. Reporting assets are in PRs #53–57; full matched analysis, both scoring policies and independent financial/observability receipts are linked from the repository. The pooled DeepSeek figures in Section 6 are exploratory, not part of the controlled factorial comparison.

References

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